

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

14th September, 2016

Subject SA4 – Pensions and Other Employee Benefits

Time allowed: Three hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific applications or work carried out from within India, mostly the APS/GNs issued by ASI/IAI, Accounting Standards by ICAI, Indian Tax and other relevant documents and funding or accounting standards issued by other bodies outside India. In view of this, it should be noted that focal point of answers is expected to be India Specific application for work generally carried out of India. However if application specific to any other country is quoted in the answer, the same you should answer the question with reference to Indian environment.*
5. *Attempt all questions in order of sequence.*
6. *Begin answer to each question on a separate sheet, however answer to sub-questions can be on the same sheet.*
7. *Mark allocations are shown in brackets.*
8. *Please check if you have received complete Question Paper and no page is missing. If so then kindly get new set of Question Paper from the Supervisor.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You are a consulting actuary in India. Your client is an Indian Multinational IT company listed both in India and UK. The primary defined benefit plan is of an unfunded Gratuity scheme that is per the minimum specified in the Payment of Gratuity Act, 1972, as amended. In addition, there is a leave accumulation encashment plan. The company also has a non-listed subsidiary registered in the UK that has 20% of its workforce deputed from its India headquartered parent company.

All India full time employees are eligible for the Gratuity and Leave benefit but only the deputed employees are eligible for leave accumulation plan. The accumulated leave days can be availed in service, with any balance at exit being encashed. The UK permanent employees only have an optional defined contribution plan that meets the necessary minimum requirements in the UK.

You are about to conduct the actuarial valuations under Accounting Standards for the Indian Parent company as at 31st December 2015 and have a planning meeting with the new company CFO. In the past, this company has reported under India GAAP as well as IFRS.

Initially, you have been asked to provide an approach paper to the CFO ahead of the meeting.

Outline the points you will make in your paper, covering the following aspects:

- i) a)** Key Purpose and Principles of accounting valuations and which accounting standards may apply for company's benefit plans (6)
- b)** List the key outline of what these standards cover (5)
- c)** Summarise the key difference areas of these standards (5)
- d)** What implications do these differences have to stakeholders (2)
- e)** Define:
 - Project Benefit Obligation;
 - Current Service Cost ;
 - Actuarial gains/Loss (2)
- ii)** At the meeting the CFO makes a comment

“My cost for the defined benefit plans should be just what I pay to people leaving the company in the year.”

What points would you make to him in your response during the meeting? (8)
- iii)** Your team has liaised with the client following the meeting and produced the following calculations for you as a summary for the stand alone India HQ legal entity employees Gratuity Plan.

Items	Amount	Notes
Project Benefit Obligation	INR 20Cr	As at 31 st December 2015
Project Benefit Obligation	INR 23Cr	As at 31 st December 2014
Current Service Cost	INR 5Cr	For the 12 month period to 31 st December 2015
Benefit payments	INR 7.5Cr	For the 12 month period to 31 st December 2015 – assumed to be made mid way through the year
Transfer Out due to Demerger	INR 5Cr	Took place on 1 st January 2015
Actuarial (gain)/loss on plan liabilities experience	INR 2.5Cr	
Actuarial (gain)/loss due to assumption changes	INR 1Cr	Discount rate changed from 8% at 31 st December 2014 to 7.25%. All other assumptions unchanged.
Duration of liabilities	7 years	

- a) What would be the difference in the P & L expense under the gratuity plan for AS15 (Revised) and IAS19 (Revised) (1)
- b) The CFO later says to you that the transfer out happened at 1st June 2015 and asks what considerations does he need to bear in mind for the PBO and P&L for the period. He also asks you to explain why the demerger transfer out amount is not a reduction in the P&L expenses. (5)
- c) Following a review of the results, the CFO calls and says the company is thinking about setting up a Gratuity Trust for the India entity. He asks you for a comprehensive paper on key considerations / risks / and implications to make this decision. (16)
- [50]

Q. 2) Back ground: A medium sized industrial company XYZ is having a contributory, funded pension scheme for its employees.

For Group A employees (Scheme A): It offers a defined contribution scheme. The company is paying at the rate of 10% pensionable pay. Following are some of the particulars of the scheme:

Total number of members covered	100
Average pensionable Pay	Rs. 1.5 lakh per month
Average age	50
Normal retirement age	60

For Group B employees (Scheme B): it offers a defined benefit scheme (Life Pension purchased from Insurance companies) providing following benefits:

On retirement (NRA 60)	1% of pensionable pay for each year of service payable from NRA. The pension is based on the average pensionable pay during the last three years of service before retirement/exit.
On death during service	Immediate pension to the spouse based on the number of years of service at the rate of 1.25% of pensionable pay at death
On resignation (Before 5 years of service)	Refund of members contribution without interest
On resignation at or after 5 years of service	Either a deferred pension is payable from NRA or an equivalent transfer value will be provided

Employees of both the Groups A & B pay 3% of pensionable pay as contribution. Further Group A employees enjoy a death in service lump sum benefit of ten times of pensionable pay at the time of death while in service. The contribution for this benefit is paid by the company in addition.

The profile of Group B employees

Age distribution	25-35	35-45	45-60
Number of members	2800	1100	800
Pensionable Pay per month per member	Rs. 20000 per month	Rs. 30000 per month	Rs. 75000 per month

There is a Scheme Actuary advising the Trustees of Scheme B who has indicated in his report that, on the valuation basis adopted, the present contribution rate together with the fund is more than sufficient to provide the benefits. Following have been the results of valuation of Scheme B during last three financial years:

	2013-14	2014-15	2015-16
Value of accrued benefits	1480 million	1530 million	1545 million
Value of assets	1510 million	1565 million	1585 million
Surplus	30 million	35 million	40 million
Recommended contribution rates	10% pensionable Pay(company) + 3% member contribution	10% pensionable Pay(company) + 3% member contribution	10% pensionable Pay(company) + 3% member contribution

HR Head of the company is interested in improvement of the benefits of employees of both the Groups A & B. The company put the proposal of HR Head in a meeting of its Board of Directors where it was decided to find out the financial implications of the proposal. You have been approached by the company to advise them on the proposals of the HR Head.

- i) Discuss on why company may want to offer improvement in benefits? (4)
- ii) The Company is considering the following proposals and as a Consultant Actuary you have been asked to illustrate (with numerical figures using sensible assumptions, wherever possible) and comment on the financial implications of each of the following proposals:

- a) Increase the accrual rate from 1% to 1.25% for group B employees (10)
- b) Offering a guaranteed pension for group A employees. The pension offered under the scheme will not be lower than 10% of pensionable pay (5)
- c) Enhancing death cover for both the groups as under:
 - Lump sum benefit of 15 times of pensionable pay for Group A employees
 - The spouse pension will be based on total service the member would have put in had he survived up to age 60 for Group B employees (7)

You may assume that you have access to the valuation reports of the Scheme Actuary. The Company proposes to use the surplus for providing the improvements. The company has also indicated, if necessary, that it may consider small increases to the company's contributions to the schemes (but not members' contribution)

Considering the uncertainty associated with the guarantee under proposal b), the Finance Director of the Company is keen to understand the cost of the guarantee.

- iii) How will you determine the cost of guarantee under proposal b) using stochastic simulations? (7)

The Trustees of the Scheme, after they came to know about the proposals of the company, have expressed concern about using the surplus for the benefit improvement as suggested by the Company. The Scheme Actuary to the Pension Scheme has been approached by the Trustees to prepare an independent report on the proposals.

- iv) Discuss the investigations that will be carried out by the Scheme Actuary on the surplus emerging from the fund over the years and the points to be included in his report on the appropriateness of using the surplus and the impact of the same on the security of benefits.

Your answer should cover:

- Reconciliation into the movement of assets & value of accrued benefits;
- Analysis of surplus (no calculations required);
- Considerations to be made by the Trustees, and
- Reporting & summary of recommendations. (Reporting part carries 4 marks) (17)

[50]
